FOR A LOCAL COMPLETE INTERSECTION MORPHISM THE SHRIEK OF THE STRUCTURE SHEAF IS INVERTIBLE AND THE SHRIEK PRESERVES PERFECT COMPLEXES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file duality.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-duality/preamble.tex is excerpted verbatim under the recipe fixed by LANE_BRIEF 22.1, as three spans, lines 8-13 (verbatim package and the reference/slogan/history environments), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0B6V (sources/stacks-duality/pinned-original-source.tex line 4909).

Allowed standard prerequisites (editorial).

- A local complete intersection morphism is perfect. More on Morphisms, Lemma labelled more-morphisms-lemma-lci-properties, cited at proof lines 4919-4921. Allows the perfect case lemma to reduce the statement.
- The perfect comparison for the shriek. Same chapter, Lemma labelled lemma-perfect-comparison-shriek, cited at proof lines 4921-4925. Reduces to the invertibility of f^!O_Y.
- Factorization of a local complete intersection morphism and the affine line case. More on Morphisms, Section labelled more-morphisms-section-lci, cited at proof lines 4927-4930, and Same chapter, Lemma labelled lemma-shriek-affine-line, cited at proof lines 4930-4934. Reduce to a Koszul regular immersion.
- The shriek of an effective Cartier divisor. Same chapter, Lemma labelled lemma-compute-for-effective-Cartier, cited at proof lines 4949-4951. Gives i^!O_X = N[1].

Source reference. Original reference 0F42: sources/stacks-duality/pinned-original-source.tex, source lines 3953–3953. Same-chapter pointer into the author environment situation at source line 3953; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 069H: sources/stacks-chow/tags.txt, source lines 8041–8041. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 4921 of the counted proof. It is

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Source reference. Original lemma 0B6U: sources/stacks-duality/pinned-original-source.tex, source lines 4798–4798. Same-chapter pointer into the author environment lemma at source line 4798; invoked at proof line 4922 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original section 068E: sources/stacks-chow/tags.txt, source lines 8003–8003. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 4927 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0AA1: sources/stacks-duality/pinned-original-source.tex, source lines 4609–4609. Same-chapter pointer into the author environment lemma at source line 4609; invoked at proof line 4929 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 09CC: sources/stacks-chow/tags.txt, source lines 11825–11825. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 4937 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0B4B: sources/stacks-duality/pinned-original-source.tex, source lines 3368–3368. Same-chapter pointer into the author environment lemma at source line 3368; invoked at proof line 4946 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *In Situation 0F42 let $f : X \to Y$ be a morphism of FTS_S. Assume $f : X \to Y$ is a local complete intersection morphism. Then*

(1) *$f^!\mathcal{O}_Y$ is an invertible object of $D(\mathcal{O}_X)$, and*
(2) *$f^!$ maps perfect complexes to perfect complexes.*

Proof. Recall that a local complete intersection morphism is perfect, see More on Morphisms, Lemma 069H. By Lemma 0B6U it suffices to show that $f^!\mathcal{O}_Y$ is an invertible object in $D(\mathcal{O}_X)$. This question is local on X and Y. Hence we may assume that $X \to Y$ factors as $X \to \mathbf{A}^n_Y \to Y$ where the first arrow is a Koszul regular immersion. See More on Morphisms, Section 068E. The result holds for $\mathbf{A}^n_Y \to Y$ by Lemma 0AA1. Thus it suffices to prove the lemma when f is a Koszul regular immersion. Working locally once again we reduce to the case $X = \mathrm{Spec}(A)$ and $Y = \mathrm{Spec}(B)$, where $A = B/(f_1, \ldots, f_r)$ for some regular sequence $f_1, \ldots, f_r \in B$ (use that for Noetherian local rings the notion of Koszul regular and regular are the same, see More on Algebra, Lemma 09CC). Thus $X \to Y$ is a composition

$$X = X_r \to X_{r-1} \to \ldots \to X_1 \to X_0 = Y$$

where each arrow is the inclusion of an effective Cartier divisor. In this way we reduce to the case of an inclusion of an effective Cartier divisor $i : D \to X$. In this case $i^!\mathcal{O}_X = \mathcal{N}[1]$ by Lemma 0B4B and the proof is complete. □